RV Educational Institutions
RV College of Engineering
Autonomous Institution Affiliated to Visvesvaraya Technological University, Belagavi
Approved by AICTE, New Delhi

DEPARTMENT OF MATHEMATICS

Course: Linear Algebra, Fourier Transforms and Statistics	**TEST-I (Scheme)**	**Maximum Marks:** 50
Course Code: MAT231AT	**Third Semester 2023-2024** **Branch: EC, EE, EI, ET**	**Time:** 10 AM to 11:30 AM **Date:** 08/01/2024

*Instructions to candidates: **Answer all questions.***

Q.No.	Question	M
1a	The following table gives production yield of wheat per hectare in some farms of a village:	

Production yield (in kg/hectare)	20-30	30-40	40-50	50-60	60-70	70-80
No. of farms	5	7	8	10	13	7

Compute second and third moments about 45.

Sol

Production yield (in kg/hectare)	No. of farms (f)	x	d=x-a	fd^2	fd^3
20-30	5	25	-20	2000	-40000
30-40	7	35	-10	700	-7000
40-50	8	45	0	0	0
50-60	10	55	10	1000	10000
60-70	13	65	20	5200	104000
70-80	7	75	30	6300	189000
	$\sum f = 50$			$\Sigma fd^2 =$ 15200	$\sum fd^3 =$ 256000

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Second moment about 45 $\mu'_2 = \frac{\Sigma fd^2}{\Sigma f} = 304$ 1

Third moment about 45 $\mu'_3 = \frac{\Sigma fd^3}{\Sigma f} = 5120$ 1

1b The first three moments of a distribution about the value 2 of the variable are 1, 16 and -40. Compute mean, variance, third moment and measure of skewness.

Sol

Given $\mu'_1 = 1,\ \mu'_2 = 16,\ \mu'_3 = 40,\ a = 2$

$\mu'_1 = \bar{x} - a \therefore \bar{x} = \mu'_1 + a = 3,\ \ \mu_2 = \mu'_2 - \mu_1'^2 = 15$ 1+1

$\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2\mu_1'^3 = \cancel{28},\ \gamma_1 = \sqrt{\beta}_1 = \frac{\mu_3}{\sqrt{\mu_2^3}} = \frac{28}{15^3} =$ ~~positive~~ly skewed. 1+1

-128088 -vely

Estimate y for given $x_1 = 10$ and $x_2 = 6$ by fitting a regression plane $y = b_0 + b_1x_1 + b_2x_2$ to the following data:

y:	90	72	54	42	30	12
x_1:	3	5	6	8	12	14
x_2:	16	10	7	4	3	2

Sol n = 6 and normal equations are

$nb_0 + b_1\sum x_1 + b_2\sum x_2 = \sum y$

$b_0\sum x_1 + b_1\sum x_1^2 + b_2\sum x_1x_2 = \sum yx_1$

$b_0\sum x_2 + b_1\sum x_1x_2 + b_2\sum x_2^2 = \sum yx_2$ 3

y	x_1	x_2	x_1^2	x_2^2	x_1x_2	x_1y	x_2y
90	3	16	9	256	48	270	1440
72	5	10	25	100	50	360	720
54	6	7	36	49	42	324	378
42	8	4	64	16	32	336	168
30	12	3	144	9	36	360	90
12	14	2	196	4	28	168	24
$\sum y$ = 300	$\sum x_1$ = 48	$\sum x_2$ = 42	$\sum x_1^2$ = 474	$\sum x_2^2$ = 434	$\sum x_1x_2$ = 236	$\sum x_1 y$ = 1818	$\sum x_2 y$ = 2820

5

$y = 61.4 - 3.6462x_1 + 2.5385x_2$. At $x_1=10, x_2=6$, $y = 40.169$ 2

5 Verify whether the following sets are subspace or not of the corresponding vector space.

(i) The subset $W = \{(x, y, z) | x - 2y + z = 0\ x, y, z \in R\}$ of the vector space of $\mathbb{R}^3$.

(ii) The subset $W = \left\{\begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix} \middle| \ x, y \in R\right\}$ of the vector space M_{2X2} (set of matrices of order $2X2$.

Sol **(i)** Let $\alpha = (x_1, y_1, z_1), \beta = (x_2, y_2, z_2) \in W, c \in \mathbb{R}$

Thefore $x_1-2y_1+z_1=0,\ x_2-2y_2+z_2=0$

(a) $\alpha + \beta = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$ s.t. $(x_1 + x_2)-2(y_1 + y_2)+ (z_1 + z_2) = 0 \therefore \alpha + \beta \in W$

(b) $c \cdot \alpha = (cx_1, cy_1, cz_1)$ s.t. $cx_1 -2cy_1+z_1 = c(x_1-2y_1+z_1)=0 \therefore c \cdot \alpha \in W$

Hence from (i) and (ii) W is a vector space of $\mathbb{R}3$. 5

(ii) Let $\alpha = \left\{\begin{bmatrix} x_1 & 0 \\ 0 & y_1 \end{bmatrix} \middle| \ x_1, y_1 \in R\right\}$ and $\beta = \left\{\begin{bmatrix} x_2 & 0 \\ 0 & y_2 \end{bmatrix} \middle| \ x_2, y_2 \in R\right\} \in W, c \in \mathbb{R}$

(a) $\alpha + \beta = \left\{\begin{bmatrix} x_1 + x_2 & 0 \\ 0 & y_1 + y_2 \end{bmatrix} \middle| \ x_1 + x_2, y_1 + y_2 \in R\right\} \therefore \alpha + \beta \in W$

(b) $c \cdot \alpha = \left\{\begin{bmatrix} cx_1 & 0 \\ 0 & cy_1 \end{bmatrix} \middle| \ cx_1, cy_1 \in R\right\} \therefore c \cdot \alpha \in W$

Hence from (i) and (ii) W is a vector space of M_{2X2}. 5

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2 The number of bacterial cells (y) per unit volume in a culture at different hours (x) is given b

x:	0	1	2	3	4	5	6
y:	43	46	82	98	123	167	199

Find the correlation coefficient between x and y. Also estimate the number of bacterial cells after 15 hours.

Sol

x	y	$X = x - \bar{x}$	$Y = y - \bar{y}$	X^2	Y^2	XY
0	43	-3	-65.286	9	4262.224	195.857
1	46	-2	-62.286	4	3879.510	124.571
2	82	-1	-26.286	1	690.939	26.286
3	98	0	-10.286	0	105.796	0
4	123	1	14.714	1	216.510	14.714
5	167	2	58.714	4	3447.367	117.429
6	199	3	90.714	9	8229.082	272.143
$\sum x=$ **21**	$\sum y=$ **758**			$\sum X_i^2=$ 28	$\sum Y_i^2=$ 20831.429	$\sum XY$=751

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n= 10, $\bar{x} = \frac{\sum x}{n} = 3$, $\bar{y} = \frac{\sum y}{n} = 108.286$, 2

$r = \frac{\sum X_i Y_i}{\sqrt{\sum X_i^2 \sum Y_i^2}} = 0.9833$, $b_{yx} = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2} = \cancel{11.73}$ 26.82 2

$y - \bar{y} = b_{yx}(x - \bar{x})$ $\therefore$ $y(x = 15) =$ $y = 26.82x + 27.82$, $\therefore$ $y(x = 15) = 430.12$ 2

3a If the correlation of coefficient between two variables x and y is 0.5 and acute angle between their lines of regression is $tan^{-1}\left(\frac{3}{5}\right)$, find the relation between standard deviations of x and y.

Sol Given $r = 0.5, tan\ \theta = 3/5$

$tan\ \theta = \frac{1-r^2}{r}\frac{\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2}$ therefore $\frac{3}{5} = \frac{1-0.25}{0.5}\frac{\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2}$ 2

$\therefore 2\sigma_x^2 + 2\sigma_y^2 - 5\sigma_x\sigma_y = 0$ and $(2\sigma_x - \sigma_y)(\sigma_x - 2\sigma_y) = 0$, $2\sigma_x = \sigma_y$ or $\sigma_x = 2\sigma_y$. 2

3b The following table gives the scores obtained by six competitors in a voice test by two judges:

Judge A	10	12	16	18	15	40
Judge B	12	18	25	25	50	25

Find the rank correlation coefficient between judge A and judge B.

Sol

Judge A (x)	Judge B (y)	R_x	R_y	$d_i = R_x - R_y$	d_i^2
10	12	6	6	0	0
12	18	5	5	0	0
16	25	3	3	0	0
18	25	2	3	-1	1
15	50	4	1	3	9
40	25	1	3	-2	4
					$\sum_{i=1}^{n} d_i^2 = 14$

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2

n= 6, m_1=3, $r_s = 1 - \frac{6\left[\sum_{i=1}^{n} d_i^2 + \frac{1}{12}(m_1^3 - m_1)\right]}{n(n^2-1)} = 0.5429$